

Multiple Choice

1. **Answer:** A

Options: A) 900; B) 441; C) 300; D) 484; E) 529

Note: Correct option: A - 900

2. **Answer:** B

Options: A) The column vectors of M form a basis for R^5 .; B) For any two distinct column vectors u and v of M, the set $\{u, v\}$ is linearly independent.; C) The trace of M is nonzero.; D) The eigenvalues of M are all nonzero.; E) The homogeneous system $Mx = 0$ has only the trivial solution.

Note: Correct option: B - For any two distinct column vectors u and v of M, the set $\{u, v\}$ is linearly independent.

3. **Answer:** D

Options: A) (0, 0); B) (6, 0); C) (9, 16); D) (0, 9); E) (3, 6)

Note: Correct option: D - (0, 9)

4. **Answer:** D

Options: A) $2x + 2x$ and $2x^2$; B) $x+x+x+x$ and x^4 ; C) $3x + 5x$ and $15x$; D) $4(2x - 6)$ and $8x - 24$; E) $3x + 3x + 3x$ and $3x^3$

Note: Correct option: D - $4(2x - 6)$ and $8x - 24$

5. **Answer:** D

Options: A) $\frac{17}{31}$; B) $\frac{33}{66}$; C) $\frac{33}{100}$; D) $\frac{17}{50}$; E) $\frac{1}{6}$

Note: Correct option: D - $\frac{17}{50}$

True / False

6. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: '6 over 7'.

7. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: '40'.

8. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

9. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

10. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: '3'.

Long Form Response

11. **Answer:** To deal with the $|x|$ term, we take cases on the sign of x : If $x \geq 0$, then we have $y^2 + 2xy + 40x = 400$. Isolating x , we have $x(2y + 40) = 400 - y^2$, which we can factor as

$$2x(y + 20) = (20 - y)(y + 20).$$

Therefore, either $y = -20$, or $2x = 20 - y$, which is equivalent to $y = 20 - 2x$. If $x < 0$, then we have $y^2 + 2xy - 40x = 400$. Again isolating x , we have $x(2y - 40) = 400 - y^2$, which we can factor as

$$2x(y - 20) = (20 - y)(y + 20).$$

Therefore, either $y = 20$, or $2x = -y - 20$, which is equivalent to $y = -20 - 2x$. Putting these four lines together, we find that the bounded region is a parallelogram with vertices at $(0, \pm 20)$, $(20, -20)$, and $(-20, 20)$, as shown below:

```
[asy]size(6cm);real f(real x) {return 20; } draw(graph(f, -25, 0)); real g(real x) { return -20; } draw(graph(g, 0, 25)); real h(real x){return 20-2*x;} draw(graph(h, 0,25)); real i(real x){return -20-2*x;} draw(graph(i, -25,0)); draw((0,-32)--(0,32),EndArrow); draw((-26,0)--(26,0),EndArrow); label("x",(26,0),S); label("y",(0,32),E); dot((0,20)--(0,-20)--(20,-20)--(-20,20));[/asy]
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 The height of the parallelogram is 40 and the base is 20, so the area of the parallelogram is $40 \cdot 20 = 800$.

Note: Students should show all steps in their mathematical solution.

12. **Answer:** 12.987. Explain why this is the correct answer and provide supporting evidence. Show the calculation or reasoning that leads to this conclusion.

Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

End of Answer Key